

PROJECT REPORT: EXAMPREP AI

Day 02 of the 21 Days of Vibecoding Challenge — Generative AI Study Companion

Author: Devaki Harish Nair (2nd Year CSE)	Interface Type: Interactive Graphical Web App (GUI)
Repository: github.com/devakihnair/21days	Core Stack: Python 3.14, Streamlit, Google GenAI SDK, PyPDF

1. Base Idea & Problem Statement

College exams and technical interviews present a major cognitive bottleneck: students are overwhelmed by 40-to-60 page slide decks, disorganized lab manuals, and dense textbook chapters. Research in cognitive psychology shows that *passive reading* yields poor long-term retention compared to **Active Recall** and **Spaced Testing**.

ExamPrep AI solves this by transforming unstructured lecture materials into a high-impact, three-pillar study hub: (1) an instant 2-minute high-yield cramming sheet, (2) digital active-recall flashcards, and (3) an interactive multiple-choice practice quiz with immediate self-grading and conceptual explanations.

2. Technology Stack & Architecture

Component	Technology	Rationale & Rationale
Frontend UI	Streamlit (v1.64)	Rapid reactive web GUI with state management, tab routing, and custom CSS.
LLM Engine	Google Gemini 2.5 Flash	Ultra-fast inference, high context window (up to 1M tokens), and structured JSON schema adherence.
SDK	google-genai (v2.25)	Official next-gen Google GenAI Python SDK.
Document Parser	pypdf (v6.19)	Robust, in-memory PDF page parsing and whitespace normalization.
Export Engine	ReportLab (v5.0)	Dynamically renders downloadable, printable summary PDF study sheets.

3. Core Workflow & Features

- A. Dual-Mode Input Pipeline:** Students can drag-and-drop any lecture PDF or paste raw notes directly into the text editor. The parser cleans artifact spaces, extracts pages, and feeds clean text into the inference engine.
- B. Structured Schema Enforcement:** Prompts instruct Gemini to return strictly typed JSON ensuring zero parsing errors when rendering cards or quiz radio buttons.
- C. Interactive Flashcard Carousel:** Features front/back state toggles, progress bars, and active recall drills with keyboard navigation.
- D. Automated Grading & Diagnostic Feedback:** The quiz module calculates accuracy percentages, reveals step-by-step reasoning for wrong answers, and launches celebratory confetti on a perfect score.
- E. Dual Export:** Summary cram sheets can be downloaded in both Markdown (.md) and formatted PDF (.pdf) for offline study.

4. Computer Science Concepts Demonstrated

1. **Structured LLM Outputs:** Forcing generative models to adhere to rigid JSON schemas for deterministic application consumption.
2. **Reactive State Machines:** Managing UI session states (`st.session_state``) across page re-runs in a stateless web runtime.
3. **In-Memory Binary Streaming:** Generating PDF study materials entirely in RAM using `io.BytesIO`` without unnecessary disk I/O.
4. **Graceful Degradation (Offline Mode):** The app seamlessly falls back to curated sample materials if an API key is absent or invalid, preventing crash states.

5. Verification & Performance

- **PDF Ingestion:** Validated on a multi-page Operating Systems lecture PDF (4,580 characters extracted in $<0.02s$).
- **Quiz Engine:** Verified state tracking, option randomization, grading accuracy, and retake resets.
- **Export Validation:** Generated and verified high-yield PDF study sheets in both binary buffer and file formats.

6. Roadmap Connection

ExamPrep AI marks the successful completion of **Day 02**. Day 03 shifts into web scraping and proactive automation with **DealDrop Alert**, a price-tracking bot that alerts students via Telegram when textbook or tech prices drop.